FINAL-YEAR PROJECT REPORT / 10 NOVEMBER 2026

AI-Powered Market Intelligence & Document Analysis using RAG + HPC on AWS

End-semester engineering status, reproducible local demonstration and measured HPC evidence.

Submission status	Evidence boundary
Local application and single-host genuine MPI verified	AWS deployment, live local LLM inference, and a licensed real-market case study remain pending.

Abstract

This project provides a local research workspace for PDF, Word and CSV market reports. It indexes source passages, retrieves evidence for questions, computes document similarity and K-Means clusters, and stores results with source IDs, input fingerprints and runtime metadata. The default workflow runs offline after installation and does not require AWS credentials, a database or paid APIs. Optional local model generation and genuine multi-process MPI execution are explicitly separated from the default sequential path.

1. Objectives and scope

Research: preserve document provenance and return inspectable excerpts rather than unsupported market claims.
Computation: derive numerical features from indexed documents and measure actual sequential or MPI runs.
Reproducibility: install from a clean ZIP on Windows or Ubuntu/WSL2 and verify with a doctor, tests and offline smoke test.
Honesty: label synthetic demonstrations, avoid simulated MPI, and state what cloud and model capabilities have not been verified.

2. Implemented pipeline

Stage	What the application does	Visible output
01 Import	Extract PDF/DOCX/CSV text, split it into source-located chunks, and retain document IDs.	Indexed library and import receipt
02 Retrieve	Use local TF-IDF ranking to return relevant source excerpts. Optional installed Ollama model can generate a grounded response.	Evidence table and page/row citations
03 Analyze	Build document-derived numerical vectors and run exact cosine top-k or Lloyd K-Means sequentially or through real MPI.	Measured runtime and process metadata
04 Inspect	Save results, source manifest and SHA-256 input fingerprint; display tables/charts and export a deterministic brief.	Results Studio, JSON and Markdown

Terminology: default retrieval is lexical TF-IDF; evidence-only output is extractive and is not LLM generation. Similarity and clusters describe document features and are not forecasts.

3. Local functional verification

The Windows Python 3.11 build passed the full automated test suite, import diagnostics, dependency checks and offline end-to-end smoke test. The smoke test creates temporary synthetic PDF/DOCX/CSV files and checks extraction, retrieval, vectorization, sequential similarity, clustering and traceable market metric calculations. The Gradio dashboard was exercised in a browser across all four tabs.

Browser action	Observed result
Upload synthetic PDF	One page indexed with source filename and page citation.
Ask a source question	Evidence-only answer returned the relevant synthetic PDF passage; no cloud or LLM call.
Run similarity with one PDF	Clear minimum-two-document guidance; no empty result is saved.
Upload second synthetic PDF	Two directed similarity matches computed; cosine score 0.60531 in this example.
Inspect and export	Newest result populated the table and chart; JSON and Markdown downloads were prepared.

The browser documents were explicitly fictional test material. Their score is an algorithm check, not a market finding. The dashboard now uses readable table headers and an informative empty Results state.

4. Genuine HPC evidence

Ubuntu under WSL2 launched four actual Open MPI ranks (0-3). Four-rank similarity and distributed K-Means matched independent sequential references on synthetic numerical inputs, including an empty partition (1, 1, 1, 0). A nine-document synthetic, document-derived dataset also completed both four-rank algorithms and retained all nine source IDs. MPI is never silently simulated when the runtime is unavailable.

Experiment	Measured result	Interpretation
3 x 16 vectors, 4 ranks	Median speedups 0.017x similarity; 0.020x K-Means	Process startup dominates tiny inputs.
20/100 documents, 2/4 ranks, 32 features	8/8 correct configurations; 0.007x-0.024x speedup	No speedup claim is justified.
9 x 384 document-derived vectors, 4 ranks	Both algorithms matched sequential references; partitions 3/2/2/2	Proves source-aware MPI execution on one host.

These short trials use two repetitions and one WSL2 host. Larger inputs, at least three repetitions and a separate multi-node experiment are needed for performance conclusions.

5. Installation and demonstration

Windows 10/11: Install Python 3.11, extract the ZIP, run `setup_windows.ps1`, then `scripts/doctor.py`, `pytest -q`, `scripts/verify_local.py`, and `run_windows.ps1` using the project virtual environment. Open `http://127.0.0.1:7860`.

Ubuntu or WSL2: Run `bash setup_linux.sh`, then the same doctor, test and smoke checks with `.venv/bin/python`, and launch with `bash run_linux.sh`. Genuine MPI requires an installed MPI runtime and the optional MPI package; verify ranks before selecting Real MPI.

Offline second laptop: Build a matching OS/Python wheelhouse on an internet-connected computer, copy it alongside the source ZIP, and use the documented `--no-index` setup mode. Once installed, the default retrieval and sequential analytics work without internet.

6. Scientific and submission limits

Item	Status at this release
Real public market case study	Not supplied. Bundled reports and browser PDFs are synthetic practice data.
Local neural and Ollama models	Optional integration exists; a live installed model was not verified.
Native Windows MPI	Launcher present; DLL blocked by Windows Application Control. WSL2 MPI worked.
AWS / Slurm / multi-node	Review templates exist; no deployment, cloud cost or multi-node run was performed.
Second physical laptop	Not available. Fresh ZIP installations were tested on Windows and WSL2 on one laptop.

7. What remains for the final academic claim

Supply and cite legally usable real reports; run a documented real-source case study; complete a larger, repeated MPI scaling study; and deploy and measure the intended AWS environment under the student account if the course requires cloud proof. Keep source files, run receipts, costs and screenshots as evidence. Do not replace missing evidence with synthetic claims.

Project references

README.md; QUICK_START.md; VERIFICATION.md; docs/EXPERIMENT_PROTOCOL.md; docs/REAL_DATA_CASE_STUDY.md. These files contain the exact commands, methods and current evidence. The repository contains no secret credentials in the release archive.